BARIS ASKIN

baskin@andrew.cmu.edu ◇ **Website** ◇ **LinkedIn** ◇ (+1) 412 816 6980 ◇ **Google Scholar**

EDUCATION

Carnegie Mellon University, Pittsburgh, USA *Fall 2022 - Spring 2027 (Expected)*
· Ph.D. in Electrical and Computer Engineering (M.Sc. en route, 2025) *Available from Feb. 2027*
Advisors: Gauri Joshi and Carlee Joe-Wong **GPA: 4.0/4.0**

Bilkent University, Ankara, Turkey *2017 - 2022*
· B.Sc. in Electrical and Electronics Engineering **GPA: 3.99/4.0**
Valedictorian – **Highest Ranked Graduate** of 2022 in the Faculty of Engineering

RESEARCH AREAS

LLMs, Interpretability, Alignment, Post-Training & RL (RLVR, distillation, fine-tuning), LLM Routing, Distributed/Federated Learning

WORK & RESEARCH EXPERIENCE

Graduate Researcher *August 2022 – Present*
Carnegie Mellon University

· LLM Alignment: emergent & subliminal misalignment analysis through training data and signal (SFT, on-/off-policy distillation) [NeurIPS'26][Preprint]
· LLM Interpretability: info.-theoretic method for internal planning horizon and branch awareness [ICLR'26]
· LLM RLVR: federated GRPO with off-policy coordination on public data [ICML'26 Workshop]
· LLM Routing: learning routers from sparse, decentralized evaluations in federated settings [NeurIPS'26]
· Federated Learning and Fine-Tuning: communication-efficient fine-tuning (LoRA) and asynchronous FL with convergence guarantees [NeurIPS'25][AISTATS'25][UAI'24]

Research Intern *May 2025 – August 2025*
NVIDIA Corporation

· Addressed client staleness in large-scale async FL, where outdated updates degrade the global model
· Developed a method to better utilize stale updates via data-free knowledge distillation [NeurIPS'26]
· My method was integrated into NVIDIA FLARE, NVIDIA's open-source FL framework [Code]

Quant Researcher Intern *June 2026 – August 2026*
IMC Trading

· Quantitative research on index options
· Analyzed large-scale trade data and developed statistical methods to detect patterns

SELECTED PUBLICATIONS

1. **Examining the Impact of Data and Training Methods on Emergent and Subliminal Misalignment**
B. Askin*, M. Ustaomeroglu*, A. Nayak*, G. Joshi, G. Qu, C. Joe-Wong
NeurIPS 2026 [**Link**]

2. **Internal Planning in Language Models: Characterizing Horizon and Branch Awareness**
M. Ustaomeroglu*, **B. Askin***, G. Joshi, C. Joe-Wong, G. Qu *equal contribution
ICLR 2026 [**Link**][**Poster**][**Video**]

3. **Reviving Stale Updates: Data-Free Knowledge Distillation for Asynchronous FL**
   B. Askin, H. R. Roth, Z. Sun, C. Joe-Wong, G. Joshi, Z. Xu
   NeurIPS 2026 [**Link**][**Code**]

4. **Federate the Router: Learning LM Routers with Sparse and Decentralized Evaluations**
   B. Askin*, S. Patel*, A. Nayak*, A. Vigano, J. Woo, G. Joshi, C. Joe-Wong
   NeurIPS 2026 [**Link**][**Code**]

5. **PubSwap: Public-Data Off-Policy Coordination for Federated RLVR**
   A. Nayak*, **B. Askin***, M. Ustaomeroglu, C. Joe-Wong, G. Joshi
   ICML 2026 Workshop on RL from World Feedback [**Link**]

6. **Federated Communication-Efficient Multi-Objective Optimization**
   B. Askin, P. Sharma, G. Joshi, C. Joe-Wong
   AISTATS 2025 [**Link**][**Code**][**Poster**]

7. **Ravan: Multi-Head Low-Rank Adaptation for Federated Fine-Tuning**
   A. Raje, **B. Askin**, D. Jhunjhunwala, G. Joshi
   NeurIPS 2025 [**Link**]

8. **FedAST: Federated Asynchronous Simultaneous Training**
   B. Askin, P. Sharma, C. Joe-Wong, G. Joshi
   UAI 2024 [**Link**][**Code**][**Poster**]

9. **Aligned Data Can Induce Misalignment via Context Confusion**
   Y. Bakman, D. N. Yaldiz, **B. Askin**, S. Roy, M. Ziyadi, S. Avestimehr, S. P. Karimireddy
   Preprint 2026 [**Link**]

10. **Federated Learning is a Lens towards a Democratized Future for the Scaling Law Era**
    H. H. Jiang, **B. Askin**, G. Joshi, C. Joe-Wong
    ICML 2026 (Position Track) [**Link**]

Undergraduate Research (Bilkent, 2020–2022): developed deep learning methods (transformers, CNNs, deep equilibrium models) for medical imaging, resulting in **2 IEEE Transactions on Medical Imaging papers** and **4 other conference/journal papers** (see **Google Scholar**).

SELECTED COURSES AND RELEVANT SKILLS

CMU: Ph.D.-level Optimization and Statistics courses, Generative AI, MDPs and RL, Deep RL
Bilkent: Linear Algebra, Stochastic Models, Signals and Systems, Digital Signal Processing
Skills: Python, PyTorch, Hugging Face Libraries, vLLM, VERL, OpenCV, Apache Spark

PROFESSIONAL SERVICE

· Reviewer for NeurIPS, ICLR, ICML, AISTATS, ISIT, AAAI, TMLR, IEEE/ACM ToN and MLSys

· Teaching Assistant for Introduction to ML for Engineers (18-661) and Algorithms for Large-scale Distributed ML and Optimization (18-667)

SELECTED AWARDS AND ADDITIONAL INFORMATION

· 2023-2024 Ben Cook Presidential Graduate Fellowship at CMU

· 2022-2023 Carnegie Institute of Technology Dean's Fellowship at CMU

· Bilkent University Comprehensive Scholarship: Full tuition waiver & stipend during the B.Sc.

· Turkish Government Merit Scholarship: Awarded stipend during the B.Sc.

· Ranked 252nd/2M in the university and 3rd/104k in the postgraduate entrance exams in Turkey

· Volunteer at Young Guru Academy (an international NGO based in Istanbul) from 2018 to 2020